

Roll No. 22001025622

61478

**B. Pharma 6th Semester
(Regular/Re-appear/Improvement)
Examination – May-2025**

BIOPHARMACEUTICS AND PHARMACOKINETICS

Paper : BP-604T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks, Attempt any *two* questions out of three from Section-B, each carry 10 marks and *seven* questions out of *nine* from Section-C, each carry 5 marks.

SECTION – A

1. Attempt all questions : $10 \times 2 = 20$

- (a) How absolute bioavailability can be calculated ?
- (b) Name any *two* protein involved in drug binding.

P. T. O.

- (c) Differentiate between carrier mediated transport and passive diffusion.
- (d) How urine pH Plays an important role in renal excretion of drugs ?
- (e) What is enterohepatic circulation ?
- (f) What is compartment model ?
- (g) Write the significance of bioequivalence studies.
- (h) What is the effect of increased V_d on loading dose ?
- (i) Define accumulation ratio.
- (j) What is Zero order elimination kinetics ?

SECTION – B

Note : Attempt any *two* questions : $2 \times 10 = 20$

2. Explain kinetics of protein binding in detail. Write a note on tissue permeability of drugs.
3. Differentiate between First order and zero order elimination kinetics. Enumerate factors causing non-linearity.
4. How does loading dose for a particular drug calculated ? Discuss kinetics of multiple dosing in detail.

SECTION – C

Note : Attempt any *seven* questions : $7 \times 5 = 35$

5. Explain in detail about plasma level time curve for a drug that follows a two compartment model system.
 6. Illustrate trapezoidal method for the calculation of AUC.
 7. How bioavailability can be measured using urinary excretion study ?
 8. Write a note on kinetics of protein binding.
 9. Define the following terms-elimination rate constant, Half-life and absorption rate constant.
 10. Discuss the significance of steady state drug levels in clinical setting.
 11. Describe briefly about nonlinear pharmacokinetics.
 12. Write a note on non-compartment model.
 13. Discuss various factors affecting renal excretion of drugs.
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Department of Pharmaceutical Sciences, I.G.U., Meerpur, Rewari
B. Pharm 6th sem, Biopharmaceutics & Pharmacokinetics
Ist sessional Exam, March 2025
(BP 604T)

Max Marks: 30

Time: 1 hr

Attempt all questions from part A (5x2)

Attempt 2 questions from part B (2x5)

Attempt 1 question from part C (1x10)

Part-A

(a). Area Under Curve (b) Volume of Distribution (c) Active transport (d) Elimination rate constant (e) Non compartment models

Part B

- (a). Discuss various factors affecting protein drug binding.
- (b). Explain Kinetics of Protein drug binding.
- (c). Briefly describe various factors affecting drug absorption

Part-C

- (a). Define the term absorption. Describe in detail mechanisms of drug absorption through GIT.
- (b). Give determination of various pharmacokinetic parameters (KE , $t_{1/2}$, V_d) in one compartment IV Bolus administration.

Department of Pharmaceutical Sciences

B. Pharm. 6th Semester, 2nd Sessional Examination, April, 2025

Paper Code – BP604T; Biopharmaceutics and Pharmacokinetics (Theory)

Total marks: 30

(5×2=10 marks)

Duration: 1 hr 30 minutes

1. Give a brief description of the following:

- a. ~~Loading and maintenance dose with formula~~ *Membrane Excretion.*
- c. Name the agents used to estimate GFR for renal function.
- d. Define 4 methods for enhancement of dissolution rate of poorly soluble drugs.
- e. USP type III & IV dissolution apparatus
- f. Pulmonary excretion of drugs

2. Write a note on the following (any two):

(2×5=10 marks)

- a. Define metabolism. Explain Phase II reactions.
- b. Describe the factors affecting renal excretion of drugs.
- c. Explain *in-vitro in-vivo* correlations with different levels.

3. Write a note on the following (any one):

(1×10=10 marks)

- a. Define bioavailability. Write its objectives. Explain different methods for measurement of bioavailability.
- b. Explain non-linear pharmacokinetics, factors causing non-linearity and Michaelis-menton method of estimating parameters.

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**B. Pharma 6th Semester
(Regular/Re-appear/Improvement)**

Examination – May, 2025

HERBAL DRUG TECHNOLOGY

Paper : BP603T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section – A is *compulsory*. Attempt any *two* questions from Section – B and any *seven* questions from Section – C.

SECTION – A

(Compulsory)

10 × 2 = 20

1. (i) Define the term "Herbal Medicine" giving suitable examples.
- (ii) What are oral hygiene products ?

P. T. O.

- (iii) Differentiate between "Asava and Aristas" giving suitable examples.
- (iv) What is the principle of Siddha system of medicine ?
- (v) What is meant by the term "Nuxraceuticals" ? Give suitable examples of marketed product.
- (vi) Name any *two* sweeteners used as herbal excipients.
- (vii) What do you mean by the term 'Patent' ?
- (viii) What is 'Herb-Drug interactions' ? Give suitable examples.
- (ix) What are the source and use of Spirulina ?
- (x) Name any *five* plant based industries involved on work on medicinal and aromatic plants.

SECTION – B**2 × 10 = 20**

2. (i) Discuss the herb-food interactions with emphasis on Hypercium and Pepper.
- (ii) Discuss the Good Manufacturing Practice of Indian system of Medicine.
- (iii) Discuss the ICH guidelines employed for the assessment of herbal drugs.

SECTION – C $7 \times 5 = 35$

3. (i) Discuss the sources, selection, identification and authentication of Herbal materials.
- (ii) Elaborate the Good agricultural practices in cultivation of medicinal plants.
- (iii) What are "Bhasma" ? Discuss its method of preparation.
- (iv) Write a note on Chicory and Ginseng and their health benefits.
- (v) What are Herbal cosmetics ? Give suitable examples.
- (vi) Elaborate Phytosomes as novel dosage forms.
- (vii) Describe the role of Schedule z of Drugs and Cosmetics Act in the regulation of manufacturing of ASU drugs.
- (viii) Write an explanatory note on Bioprospecting and Biopiracy.
- (ix) Enlist various recipients used in herbal formulations.
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IGU, Meerpur, Rewari

B. Pharm, 6th Sem, Herbal Drug Technology

Time: 1.5hr

MM: 30

SECTION-A

Attempt all questions

2*5=10

1. What is the basic principle of homeopathy?
2. Give general method of preparation of bhasamas.
3. Differentiate between aristas and asavas.
4. Give side effects and interactions of pepper.
5. What is the role of nutraceuticals in GIT diseases.

SECTION-B (Attempt any 2)

5*2=10

6. How will you select, identify and authenticate herbal materials.
7. Write brief note on pest management in medicinal plants.
8. Discuss the role of nutraceuticals with suitable examples in Diabetes/CVS disorders.

SECTION-C (Attempt any 1)

10*1=10

9. Discuss herb-drug and herb-food interaction. Also, study possible side-effects of garlic and hypericum.
10. Elaborate good agricultural practices.

Indira Gandhi University
Department of Pharmaceutical Sciences
B. Pharm. 6th Semester, 2nd Sessional Examination, 23 April 2025
Paper Code – BPH 603T; Herbal Drug Technology

Total marks: 30

Duration: 1 hr 30 min.

Q1 is compulsory. Each part carries 2 marks

Explain in detail

- i) IPR
- ii) Excipients
- iii) Bioprospecting
- iv) Farmer's right
- v) ASU DCC

Attempt any two out of three. Each question carries 5 marks

- Q2. Write about WHO guidelines for the assessment of herbal drugs.
- Q3. Write a note on herbal sweeteners and binders.
- Q4. Define Colorants. Give its classification.

Attempt any one out of two. Each question carries 10 marks

- Q5. Write a detailed note on patenting of traditional knowledge and natural products with examples.
- Q6. Explain traditional herbal formulations in detail.

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**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)
Examination – May-2025**

MEDICINAL CHEMISTRY III

Paper : BP601T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *all* questions from Section-A & any *two* questions from Section-B and any *Seven* questions from Section-C.

SECTION – A

1. Attempt *all* questions : 2 × 10 = 20
- (a) What are Cephalosporins.
 - (b) Draw the structure of Amantadine and give its use.
 - (c) Name the agents used in the treatment of TB.

P. T. O.

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- (d) Draw structure of Chloroquin and give its use.
- (e) Introduce Penicillins with *two* examples.
- (f) Write structure and use of Sulphacetamide.
- (g) Define Prodrugs ?
- (h) What are antimalarial agents ?
- (i) Define Beta Lactam Antibiotics.
- (j) What are Antiprotozoal drugs ? Give any two examples.

SECTION – B

2. Long Answer Questions (Attempt any *two*) : $10 \times 2 = 20$

- (a) Explain in detail about QSAR.
- (b) Discuss the SAR quinolines, MOA and uses.
- (c) Explain Application of Prodrugs.

SECTION – C

3. Short Answer Questions (Attempt any *Seven*) :

$$7 \times 5 = 35$$

- (a) Discuss the SAR of Tetracyclines with suitable examples.
- (b) Classify anti-tubercular agents.

- (c) Discuss the chemical structure MOA and synthesis of Isoniazid.
 - (d) Describe Anthelmintics giving examples.
 - (e) Discuss in detail about anti- TB agents.
 - (f) Write a note on Docking techniques.
 - (g) Write the mechanism of action and synthesis of Ciprofloxacin.
 - (h) Write a note on Combinatorial Chemistry.
 - (i) Write the mechanism of action and synthesis of Sulphamethoxazole.
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DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI

First Sessional Theory Examination of B. Pharmacy 6th Semester –February, 2025

Subject: Medicinal Chemistry - III Theory (Paper Code: BP601T)

(Exam Date: 27.02.2025)

Time: 1 hr

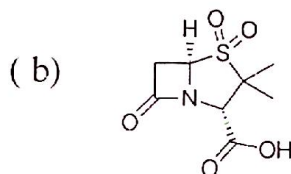
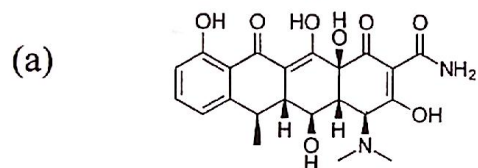
Max. Marks =30

Attempt all questions

Q1. Give the answers of all questions -

5 × 2 =10

- i. Define antibiotic with example
- ii. Sketch the molecular structure of β -Lactam ring with IUPAC name.
- iii. Identify the name of given structures
- vi. Write down the two examples of Macrolide antibiotics
- v. Write the synthesis of Chloramphenicol.



Q2. Attempt any two question (Short answer type questions)

2 × 5 = 10

- a) Discuss the history of antibiotic with suitable examples.
- b) Discuss the structure activity relationship of quinolones.
- c) What is the degradation? Discuss the degradation of penicillin.

Q3. Attempt any one question (Long answer type questions)

1 × 10 = 10

- a) What is the tetracycline? Discuss the mechanism and nomenclature of tetracycline with suitable drug example.
- b) What are the prodrugs? Discuss the type and applications of prodrug design.

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DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI

Second Sessional Theory Examination of B. Pharmacy 6th Semester –April, 2025

Subject: Medicinal Chemistry - III Theory (Paper Code: BP601T)

(Exam Date: 21.04.2025)

Max. Marks =30

Time: 1 hr

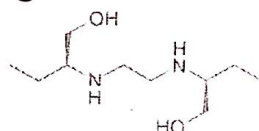
Attempt all questions

Q1. Give the answers of all questions -

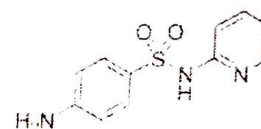
5 × 2 =10

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|---|--|
| i. Write the uses of Sparfloxacin and Tolnaftate drugs. | iv Write down the mechanism action of Antifungal agents. |
| ii. Sketch the molecular structure of Cotrimoxazole. | v Write the synthesis of Nitrofurantoin drug. |
| iii. Write the name of given structures | |

(a)



(b)



Q2. Attempt any two question (Short answer type questions)

2 × 5 = 10

- Discuss the classification of antiviral agents with molecular structure.
- Discuss the structure activity relationship of Sulphonamides with suitable examples.
- Write the synthesis of Isoniazid and Dapsone drugs.

Q3. Attempt any one question (Long answer type questions)

1 × 10 = 10

- Discuss the drug profile (including about the molecular structure, mechanism of action, and uses) of Pyrazinamide, Ofloxacin, Zidovudine and Thiabendazole drugs.
- Write the synthesis of Mebendazole and Ciprofloxacin drugs in details

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**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)
Examination – May-2025**

QUALITY ASSURANCE

Paper : BP606T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-I is *compulsory*. Attempt any *two* out of *three* question from Section-II and any *seven* out of *nine* questions from Section III.

SECTION – I

1. (a) Define Total Quality Management. 20 × 1 = 20
(b) Define contamination.
(c) Define quality assurance.
(d) Write full form of CTD and SOP.

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- (e) What is DMF.
- (f) Define validation.
- (g) Name any four types of closures.
- (h) What is 'Recall' in pharmaceutical industry ?
- (i) Define GLP.
- (j) Define FIFO.
- (k) What are the records ?
- (l) Define quality control.
- (m) Define calibration.
- (n) What is master formula record ?
- (o) Define QbD.
- (p) Write down the main role of quality documentation in pharmaceutical industry.
- (q) Give any *two* application of Q-series guidelines.
- (r) Discuss the chief role of ICH.
- (s) What is GMP ?
- (t) What are the Personal responsibilities ?

SECTION – II

2. Explain the design, construction, plant layout and requirement of environmental control in sterile manufacturing unit. 10
3. Write in detail about NABL accreditation, principles and procedures. 10
4. Write the quality control test for containers and glass apparatus. 10

SECTION – III

5. Write a short note on Q series guidelines of ICH. 5
6. Discuss master formula record. 5
7. Write a brief note on QbD tools. 5
8. Write down the main objectives of ICH. 5
9. Differentiate between validation and calibration. 5
10. Write a short note on quality audit. 5
11. Write the importance of quality documentation. 5
12. Enlist the merits of warehousing. 5
13. Write a short note on ISO 9000. 5

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B.Pharma 6th Semester, 1st Sessional Examination, 2025

Paper Code- BP606T Pharmaceutical Quality Assurance

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

a. Define Qa and Qc

2x5=10 marks

b. Name of elements of Qbd

c. Define GMP?

d. Define benefits of ISO

e. Define Participants of ICH

2 Attempt any one question from the rest

10x1=10 marks

1.a. Briefly Explain about ICH

2.b. Briefly explain about GLP practices

3 Attempt any two questions from the rest

5x2=10 marks

a. Elements and registration process of ISO

b. Principle and procedures of NABL accreditation

c. Quality control test for container

B. Pharmacy 6th Semester 2nd Sessional Exam 2025
Pharmaceutical Quality Assurance BP606T

Time: 1.5 Hrs

Max. Marks: 30

SECTION-A

Attempt all Questions. Each question carry 2 Marks.

2X5=10

1. Define Corrective action & Preventive action.
2. What is retrospective validation?
3. Define Master formula record.
4. Qualification & Validation.
5. Define quality audit.

SECTION-B

Short Answer Questions. Attempt any 2 out of Three.

5X2=10

6. What steps involved in handling of complaints.
7. Explain batch formula Record.
8. Define complaints and explain types of Complaints.

SECTION-C

Long Answer type Question. Attempt any One out of Two

10X1=10

9. Give brief account of documentation in Pharmaceutical Industry.
10. Outline the procedure for material warehousing in Pharmaceutical Industry.

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**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)**

Examination – May, 2025

PHARMACEUTICAL BIOTECHNOLOGY

Paper : BP605T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory* consisting of ten questions carrying 2 marks each. Section-B contains nine questions carrying 5 marks each and students have to attempt any *seven* questions. Section-C contains *three* questions carrying 10 marks each and students have to attempt any *two* questions.

SECTION – A

1. Attempt *all* questions.

10 × 2 = 20

(a) What is Protein Engineering ?

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- (b) Enlist some applications of biosensors in pharmaceutical industries.
- (c) Enumerate uses of DNA ligase and restriction endonucleases.
- (d) Describe the significance of multiple cloning sites in cloning vector.
- (e) What are hypersensitivity reactions ?
- (f) Write down the applications of hybridoma technology.
- (g) Write the principle of Sandwich ELISA.
- (h) What is frameshift mutation ?
- (i) Enlist *four* equipments used in fermentation.
- (j) Enumerate uses of dried human plasma.

SECTION – B

Note : Attempt any *seven* questions $5 \times 7 = 35$

- 2. Discuss basic principles of genetic engineering.
- 3. Describe any *two* methods of enzyme immobilization.
- 4. Explain the production and industrial application of Lipase.
- 5. Write an exhaustive note on recombinant DNA technology.

6. Discuss the importance of PCR technique.
7. Describe method of preparation of toxoids.
8. Write a short note on structure and function of MHC.
9. Differentiate genome organization of Eukaryotes and Prokaryotes.
10. Write a note on collection, processing and storage of whole human blood.

SECTION – C

Note : Attempt any *two* question :

2 × 10 = 20

11. Discuss principle, various steps and applications of western blotting.
12. Discuss the production of penicillins by fermentation.
13. Write short notes on any *two* :
 - (i) Aeration process in fermentation
 - (ii) Plasma substitutes
 - (iii) Structure of immunoglobulins

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharma 6th sem 1st Sessional Examination, 2025
Paper Code- BP605T, Pharmaceutical Biotechnology

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

- a. Define Biotechnology to Pharmaceutical science $2 \times 5 = 10$
- b. Define Enzyme immobilization and one application
- c. Principle of genetic engineering and protein engineering
- d. Define Working of biosensors and application
- e. Define cloning vectors

2 Attempt any one questions from the rest $10 \times 1 = 10$ marks

- 1.a. Describe Brief introduction to PCR
- 2.b. application of r-Dna technology and production of genetic engineering 1.vaccines-hepatitis and interferon

3 Attempt any two questions from the rest $5 \times 2 = 10$ marks

- a. Explain Biosensors working and application in pharmaceutical industries
- b. Explain Method of enzyme immobilization and application
- c. Explain restriction endonucleases with example

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**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)**

Examination – May, 2025

PHARMACOLOGY - III

Paper : BP602T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks. Attempt any *seven* questions out of *nine* from Section-B, each carry 5 marks and *two* questions out of *three* from Section-C, each carry 10 marks.

SECTION – A

Note : Attempt *all* questions.

$2 \times 10 = 20$

1. (a) Describe the mechanism of action of Bromhexine as mucolytic agent.
- (b) Write a short note on Montelukast.
- (c) Define Cross- resistance.

P. T. O.

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- (d) How aminoglycosides produce adverse effects such as ototoxicity and nephrotoxicity ?
- (e) Describe the mechanism of action of Isoniazid.
- (f) Enlist drugs used for the treatment of Hepatitis C.
- (g) Write the mechanism of action of Cyclophosphamide.
- (h) What are Calcineurin inhibitors ?
- (i) Enumerate clinical symptoms of Morphine poisoning.
- (j) Differentiate between genotoxicity and mutagenicity.

SECTION – B

Note : Attempt any *seven* questions.

5 × 7 = 35

2. Classify gastric acid secretion inhibitors.
3. Discuss various drugs used in the treatment of Helicobacter pylori infection.
4. Write a note on drugs used in the management of COPD.
5. Illustrate the mechanism of action and adverse effects of Tetracycline and Levofloxacin.
6. Write down the classification of Penicillins.

7. Discuss the mechanism of action and adverse effects of Ethambutol and Rifampicin.
8. Discuss the clinical symptoms and management of mercury poisoning.
9. Discuss the mechanism of action of any *three* drugs clinically useful in sexually transmitted diseases.
10. Write short notes on Albendazole and Niclosamide.

SECTION – C

Note : Attempt any *two* questions.

2 × 10 = 20

11. Discuss various 2nd line of anti-TB drugs. Write the mechanism of action of Primaquine and Atovaquone.
 12. Write down the classification of antiretroviral drugs. Explain the mechanism of action of Ganciclovir.
 13. Classify hormonal drugs used as anti-cancer drugs. Describe the importance of biosimilars.
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Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharmacy 6th Semester Second Sessional Examination
Pharmacology III (Paper code- BP602T)

Date- 22nd April, 2025

Duration- 1h 30 min

Max. Marks- 30

Question 1 is compulsory. Each part carries 2 marks.

Q1. Write a note on the following-

- a. Constipation and diarrhea
- b. Appetite stimulants
- c. Digestants
- d. ~~Biological clock~~ *Mutagenicity*
- e. Emetics and Anti emetics

Attempt any two out of three. Each question carries 5 marks.

Q2. Briefly write about acute, subacute and chronic toxicity studies

Q3. Classify anti-fungal drugs with examples. Write about azoles

Q4. Classify antimalarial drugs. Write the pharmacology of any one class of drugs.

Attempt any one out of two. Each question carries 10 marks.

Q5. Classify antitubercular agents and write the MOA and adverse effects of isoniazid.

Q6. Classify antiviral drugs with examples. Write the mechanism of action, adverse drug reaction and uses of NRTI'S.

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharma 6th sem 1st Sessional Examination, 2025
Paper Code- BP602 T. PHARMACOLOGY-III

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

2x5=10 marks

- Define chemotherapy.?
- What is the difference between a productive and non-productive cough?
- What are the common side effects of aminoglycosides?
- What is the mechanism of action of corticosteroids in asthma?
- Define Antileprotic agents?

2. Attempt anyone questions from the rest...

10x1=10 marks

- Describe in detail the classification, mechanism of action, and side effects of expectorants and antitussives.
- Discuss antifungal agents with their mechanism of action, classification, and therapeutic uses

5x2=10 marks

3. Attempt any two questions from the rest...

- Describe the classification of anti-asthmatic drugs with examples.
- Describe the mechanism of action and uses of cotrimoxazole.
- Write a note on antitubercular agents?